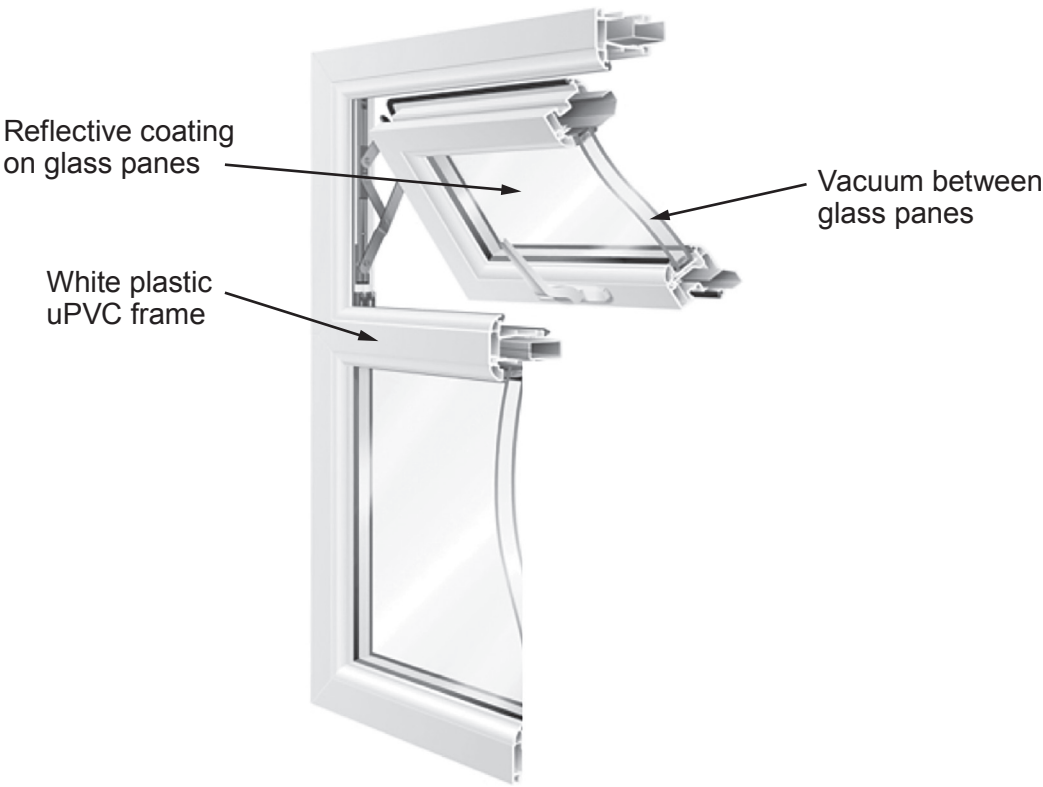


5. (a) A householder decides to replace his single glazed, aluminium framed windows. A company recommends fitting 'Energy Windows' like the one shown below.



Explain in terms of conduction, convection and radiation how the labelled features will help to reduce energy losses from the house. [6 QER]

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(b) A family fits double glazing to their house at a cost of £4 000. They save £80 per year on their heating bill.

(i) Calculate the payback time. [1]

Payback time = years

(ii) Suggest **two** reasons why this payback time may change. [2]

1.

2.

9

